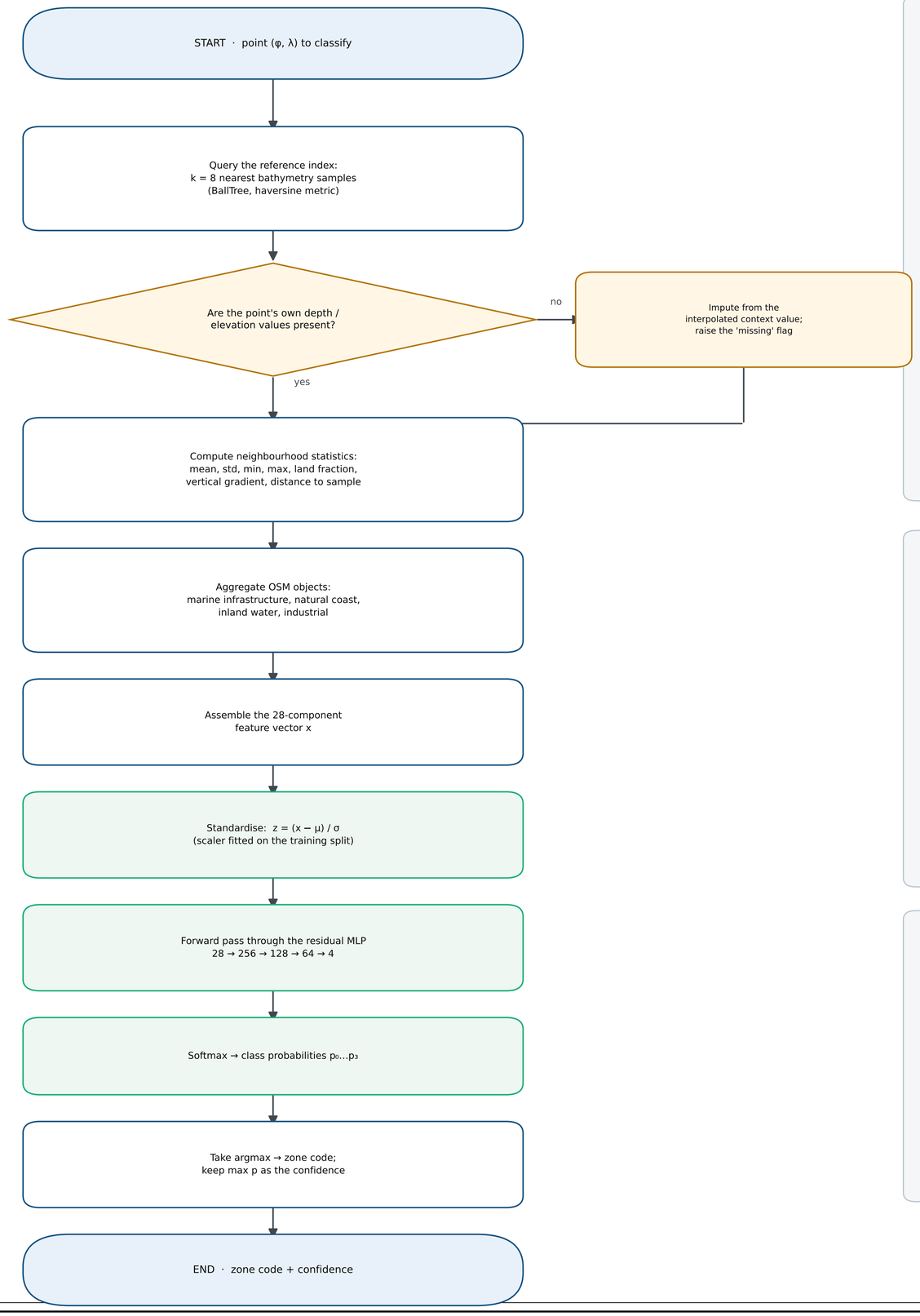
Algorithm of geographic-zone classification

Flowchart of the neural-network classification procedure

Sheet 4

Format A1



Feature vector (28 components)

Geodetic encoding — 4

sin φ, cos φ, sin λ, cos λ (continuous across the antimeridian, no artificial discontinuity)

Point measurements — 5

signed-log depth, missing flag, height/depth flag, signed-log elevation, missing flag

Neighbourhood context — 9

mean / std / min / max depth of the 8 nearest samples, range, land fraction, sign agreement, log distance to the nearest sample, gradient

OSM objects — 10

marine infrastructure, natural coastline, inland water, industrial — each at the point and averaged over the neighbourhood; plus totals

Why a reference index is not leakage

- The index stores only depth, elevation and OSM counters — never a class label.
- It is built from the training split alone; validation and test points are never inserted.
- During training a point never queries itself (exclude_self = True).
- In deployment the same role is played by a NOAA bathymetry raster and an OSM extract, so the feature is reproducible for any coordinate on Earth.

Achieved quality

Class	Precision	Recall	F1	Support
OPEN_SEA	0.9835	0.9639	0.9736	8,348
COASTAL_SEA	0.9433	0.9721	0.9575	5,548
NEAR_COAST	0.9703	0.9709	0.9706	4,537
COASTLINE	0.9768	0.9755	0.9761	4,279
Overall accuracy			0.9695	22,712

Application for constructing a maritime vessel's course based on neural-network classification of geographic zones

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